Edge Reflections and Periodic Slopes in Lattice Billiards

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Related Lean formalization: a conditional formalization of the verified core is provided; the remaining geometric ingredients are isolated as explicit external inputs.

Abstract

Let P be a compact connected polygonal billiard table whose directional reflection group is finite, and let (X_P, ω_P) be its reduced translation-surface unfolding. We show that the direction of every edge of P contains either a saddle connection or a regular closed geodesic on X_P. The key point is that the reflection fixing the direction of an edge acts on the reduced unfolding and fixes a regular segment coming from the interior of that edge.

If P is a lattice polygon, the Veech dichotomy implies that every edge direction is completely periodic and parabolic. Consequently every nonsingular billiard trajectory parallel to an edge is periodic. In particular, this gives an affirmative answer to Question 3.13 of McMullen, which asks whether every edge of a lattice polygon has periodic slope.

We also record the additional cusp structure supplied by the edge reflection. Every edge cusp is fixed by an orientation-reversing element of the extended Veech group, and its projective extended stabilizer is infinite dihedral. If the directional reflection group is D_N, then the edge cusps form one Veech-group orbit when N is odd, and one or two orbits when N is even.

Feedback. This note has not undergone independent human mathematical review. Readers who know relevant prior literature, identify an error, or wish to check the argument are especially encouraged to contact Chuyang Chen at the address listed at the end of the paper or to open an issue in the repository.

1 Introduction

Polygonal billiards are a classical source of translation surfaces. For a rational billiard table P, successive reflections of a billiard trajectory can be unfolded to a straight trajectory on a compact translation surface. When the associated Veech group is a lattice, the resulting directional dynamics is especially rigid.

McMullen asks whether this rigidity forces every actual edge of a lattice polygon to have periodic slope [**?**, Question 3.13]. A positive answer was previously known when the trace field is quadratic [**?**, Corollary 6.3]. The question is subtle because a connection between two vertices need not force periodicity: McMullen exhibits a lattice triangle of type $(2, 5, 7)$ with a vertex connection parallel to an aperiodic trajectory [**?**, §3].

The distinction for an actual edge is global rather than local. An edge is not merely a geodesic segment joining two vertices: on the unfolding it is a mirror for a reflection symmetry. The main issue is to place that reflection on the correct translation surface. On the classical full reflection unfolding the symmetry is manifest, but the lattice condition is imposed on the reduced unfolding. These surfaces can be genuinely different. For example, for the regular hexagon McMullen's full surface has genus 4 while the reduced surface has genus 1, and the map between them is a degree-6 branched covering [**?**, Appendix A].

We use McMullen's explicit equivariant description of the reduction [**?**, Appendix A] to make this compatibility transparent. If G is the finite directional reflection group and H is the subgroup responsible for translation redundancy, the full surface Y carries commuting actions of G and H, and the reduced surface is

$$X_P \cong Y/H.$$

Consequently the G-action itself descends to X_P. The defining side gluing then shows that the reflection associated with an edge fixes the image of the interior of that edge pointwise.

The dynamical consequence of such a fixed mirror is elementary and is already implicit in Vorobets' work on affine reflections. In the proof of [**?**, Proposition 2.1], Vorobets considers the trajectory through a fixed point in the $+1$-eigendirection of a reflection and observes that it is fixed pointwise and finite. We record the corresponding translation-surface statement in a self-contained form below.

Thus neither the fixed-axis observation nor the equivariant unfolding formalism is claimed here as a new general construction. The point is the compatibility between them on the particular reduced surface used to formulate the lattice condition: an actual edge still supplies a fixed mirror there. We have not found this consequence stated in the earlier literature; in particular, the current author version of McMullen's survey still asks whether every edge of a lattice polygon has periodic slope [**?**, Question 3.13].

Our first result does not require the lattice condition.

Theorem 1.1 (Edge directions on rational billiards)**.** *Let P be a compact connected polygonal billiard table whose directional reflection group is finite. Let (X_P, ω_P) be its reduced unfolding. For every edge e of P, the direction parallel to e contains either a saddle connection or a regular closed geodesic on X_P. In particular, every edge direction is nonminimal.*

For lattice polygons, Veech's theorem immediately upgrades this to complete periodicity.

Theorem 1.2 (Periodic edge directions)**.** *Let P be a lattice polygon. Every edge direction of P is completely periodic and parabolic on (X_P, ω_P). Moreover every nonsingular billiard trajectory parallel to an edge of P is periodic.*

Corollary 1.3. *Every edge of a lattice polygon has periodic slope. Thus [**?**, Question 3.13] has an affirmative answer.*

The orientation-reversing character of the reflection is useful rather than problematic. The reflection lies in the extended affine group and detects a distinguished nonminimal direction; the ordinary orientation-preserving Veech group then supplies the periodic/minimal dichotomy. Schematically,

orientation-reversing reflection	orientation-preserving lattice group
$\Downarrow$	$\Downarrow$
fixed edge mirror	Veech dichotomy
$\Downarrow$	$\Downarrow$
nonminimal direction	complete periodicity.

After the proof of Corollary **??**, we record two short consequences of the same reflection symmetry. These are not needed for the answer to Question 3.13, but describe the cusp data naturally marked by the polygonal presentation. The edge reflection fixes that cusp, and the projective extended cusp stabilizer is infinite dihedral. The finite dihedral group $G(P)$ also strongly restricts the number of cusp classes represented by actual edges: there is one such class when the rotation order is odd and one or two when it is even. The regular octagon and the $(\pi/8, 3\pi/8, \pi/2)$ right triangle illustrate how this edge-cusp data can depend on the billiard presentation even when the reduced translation surface is the same.

The paper is organized as follows. Section 2 recalls the full and reduced unfoldings and proves that an edge remains a fixed mirror on the reduced surface. Section 3 gives the fixed-axis argument and proves Theorem **??**. Section 4 applies the Veech dichotomy and proves Theorem **??** and Corollary **??**. Section 5 studies the resulting reflective cusps, and Section 6 discusses their dihedral orbit structure and examples.

2 Full and reduced unfoldings

By a polygonal billiard table we mean the closure of a bounded connected planar domain whose boundary is a finite disjoint union of polygonal Jordan curves. Thus holes are allowed. Let $P \subset \mathbb{R}^2$ be such a table. For every edge $e \in E(P)$, let

$$r_e \in O(2)$$

be the linear reflection whose fixed line is parallel to e, and set

$$G(P) = \langle r_e : e \in E(P) \rangle \subset O(2).$$

For the purposes of this paper we call P rational when $G(P)$ is finite. For an ordinary simply connected polygon this is equivalent to requiring all interior angles to be rational multiples of π; the group-theoretic formulation is the one appropriate when several boundary components are allowed. See [**?**, §1] for the standard reflection-group formulation of rational billiards.

We use the explicit equivariant model of the full and reduced unfoldings from [**?**, Appendix A]. Translate P so that its barycenter is the origin and put

$$G = G(P), \qquad H = \{h \in G : hP = P\}.$$

Before this normalization, H is equivalently the subgroup preserving P up to translation.

The full reflection unfolding is

$$(Y, \eta) = \left(\bigsqcup_{g \in G} (gP, dz) \right) / \sim,$$

where an edge a of gP is identified with $r_a a$ in $r_a gP$, and r_a is the unique element of G reflecting in the direction of a. The reduced unfolding is

$$(X_P, \omega_P) = \left(\bigsqcup_{[g] \in G/H} (gP, dz) \right) / \sim .$$

McMullen constructs an isometric action

$$G \times H \longrightarrow \mathrm{Isom}(Y, |\eta|)$$

whose two factors commute, with derivatives

$$D\psi_g = g, \qquad D\phi_h = I.$$

Moreover,

$$X_P \cong Y/(\{1\} \times H), \qquad P \cong Y/(G \times \{1\}),$$

and the G-action descends to X_P [**?**, Appendix A, equation (A.3)]. We denote the descended action again by $g \cdot x$.

Let

$$\iota_Y : P \longrightarrow Y$$

be the inclusion of the identity copy and let

$$\chi : Y \longrightarrow X_P$$

be the reduction map. Set

$$\iota_X = \chi \circ \iota_Y : P \longrightarrow X_P.$$

The map ι_X need not be injective on ∂P, but it is locally isometric along edge interiors. Indeed, McMullen proves that χ is a regular branched covering and that its critical points lie over specified vertices of P [**?**, Theorem A.6].

The following is the point at which the reflection symmetry meets the reduced surface.

Lemma 2.1 (Edges remain fixed mirrors)**.** *Let e be an edge of P. The element $r_e \in G$ induces an orientation-reversing affine involution*

$$\sigma_e : X_P \longrightarrow X_P$$

with

$$D\sigma_e = r_e.$$

For every $x \in e^\circ$, a sufficiently small interval $J \subset e^\circ$ containing x maps isometrically to a nondegenerate regular geodesic segment

$$\iota_X(J) \subset \mathrm{Fix}(\sigma_e).$$

Proof. On the full unfolding, the defining edge gluing identifies the point x in the identity copy with the corresponding point $r_e x$ in the copy $r_e P$. But the left G-action by r_e sends the first representative precisely to the second. Hence

$$\psi_{r_e}(\iota_Y(x)) = \iota_Y(x)$$

as points of Y. Since the G-action commutes with the H-action, it descends to X_P, and therefore

$$\begin{aligned} \sigma_e(\iota_X(x)) &= \sigma_e(\chi(\iota_Y(x))) \\ &= \chi(\psi_{r_e}(\iota_Y(x))) \\ &= \chi(\iota_Y(x)) = \iota_X(x). \end{aligned}$$

Thus the image of e° is fixed pointwise.

Finally x is not a vertex, so $\iota_Y(x)$ is not a critical point of χ. By [**?**, Theorem A.6], χ is a local isometry in a neighborhood of $\iota_Y(x)$. After shrinking to a small interval $J \subset e^\circ$, the image $\iota_X(J)$ is therefore a nondegenerate regular geodesic segment fixed pointwise by σ_e. □

Remark 2.2 (Antecedents). Reflection symmetries of the classical unfolding and their dynamical use go back to the standard theory of rational billiards; Vorobets' argument in [**?**, Proposition 2.1] contains the fixed-axis observation used in Section 3. Lemma **??** is the elementary compatibility that places this mirror on McMullen's reduced surface.

Remark 2.3 (Why the reduction matters). The fixed-mirror statement is immediate on the full reflection unfolding. What matters for the present problem is that it survives on the reduced surface on which the lattice condition is imposed. The reduction can be substantial: for the regular hexagon, for example, the reduced surface has genus 1, the full surface has genus 4, and the map between them is a degree-6 regular branched covering [**?**, Appendix A]. In particular, its branch values are regular points on the torus. As discussed in [**?**, §5.4], marking or forgetting such regular points can change the affine and Veech groups. The argument above avoids any transfer of the lattice hypothesis through this cover: the reflection itself descends.

3 Reflection axes and edge directions

We isolate the elementary consequence of a fixed mirror. A closely related observation occurs in the proof of [**?**, Proposition 2.1].

Lemma 3.1 (Fixed-axis lemma)**.** *Let (X, ω) be a compact translation surface and let $\sigma : X \to X$ be an affine involution whose derivative is a Euclidean reflection. Let θ be the $+1$-eigendirection of $D\sigma$. If $\mathrm{Fix}(\sigma)$ contains a regular point, then the direction θ contains either a saddle connection or a regular closed geodesic.*

Proof. Because $D\sigma$ is a Euclidean reflection, σ is antiholomorphic on the regular part of the translation surface and extends antiholomorphically across the zeros of ω. Near any fixed point, a local holomorphic coordinate can be chosen in which σ is complex conjugation. Hence, when nonempty, $\mathrm{Fix}(\sigma)$ is a smooth compact real one-dimensional submanifold of the underlying compact Riemann surface, and therefore a finite disjoint union of circles.

Let C be the component of $\mathrm{Fix}(\sigma)$ containing the given regular point. On $C \setminus Z(\omega)$, the tangent line is the $+1$-eigenspace of $D\sigma$, so every regular arc of C is a straight geodesic in direction θ.

If $C \cap Z(\omega) = \varnothing$, then C itself is a regular closed geodesic. If $C \cap Z(\omega) \neq \varnothing$, the finite set $C \cap Z(\omega)$ cuts the circle C into finitely many open geodesic arcs. The closure of each such arc is a geodesic segment whose interior contains no zero and whose endpoints are zeros of ω; the two endpoints are allowed to coincide. Thus its closure is a saddle connection. □

Remark 3.2 (Vorobets' two eigendirections). Vorobets' stated "symmetric trajectory" uses the -1-eigendirection of a reflection, which in the billiard picture is normal to the reflecting side. His Proposition 2.1 shows that a trajectory through a nonsingular fixed point in this direction is periodic or a saddle link. In the same proof he simultaneously considers the $+1$-eigendirection through the fixed point and observes that this trajectory is fixed pointwise and finite [**?**, Proposition 2.1]. The latter is the direction parallel to an edge and is the part relevant to McMullen's question.

Applied to the edge reflection σ_e on the reduced surface, these two observations show that both eigendirections of $D\sigma_e$ contain a finite geodesic: the $+1$-direction is parallel to e, while the -1-direction is perpendicular to e. The perpendicular billiard application is classical and is stated explicitly by Vorobets; our main use is the parallel direction.

Proof of Theorem **??**. Let e be an edge of P. By Lemma **??**, the affine involution σ_e has derivative r_e and has a regular fixed point lying on a fixed geodesic segment parallel to e. Apply Lemma **??**. The edge direction therefore contains either a saddle connection or a regular closed geodesic. In either case the directional foliation is not minimal. □

Remark 3.3 (Edges versus vertex connections). A general vertex connection carries only local geodesic information and need not determine a periodic direction. The additional structure of an actual edge is the global relation

$$\iota_X(e^\circ) \subset \mathrm{Fix}(\sigma_e),$$

which forces its direction to be nonminimal.

4 Lattice polygons

Assume now that P is a lattice polygon, so that (X_P, ω_P) is a Veech surface.

We use the topological form of the Veech dichotomy as stated in [**?**, Theorem 5.10], originating in Veech's work [**?**]. On a lattice surface, every direction is either minimal or every leaf is closed or a saddle connection. In the latter case the surface decomposes into cylinders and a nontrivial affine multitwist preserves the decomposition. In particular, every nonminimal direction is completely periodic and parabolic.

Proof of Theorem **??**. By Theorem **??**, every edge direction is nonminimal on (X_P, ω_P). The Veech dichotomy therefore makes every edge direction completely periodic; the affine multitwist supplied by [**?**, Theorem 5.10] shows that it is parabolic.

McMullen's billiard form of the Veech dichotomy says that, for a lattice polygon, billiard trajectories of a fixed slope fall into the periodic alternative or the uniformly distributed alternative [**?**, Theorem 3.2]. The completely periodic cylinder decomposition places the edge slope in the periodic alternative. Hence every nonsingular billiard trajectory parallel to the edge is periodic. □

Proof of Corollary **??**. Apply Theorem **??** to each edge. □

Remark 4.1 (A stronger conclusion). McMullen's question asks only whether an edge has periodic slope. The argument gives the stronger statement that the entire nonsingular edge-parallel billiard flow is periodic.

5 Reflective cusps

The main result is complete at this point. We finish by recording two short consequences of the same reflection symmetry, first for cusp stabilizers and then for the dihedral organization of edge directions.

Let P be a lattice polygon and abbreviate

$$(X, \omega) = (X_P, \omega_P).$$

Write

$$\Gamma = \mathbb{P}D\,\mathrm{Aff}^+(X, \omega) \subset \mathrm{PSL}_2(\mathbb{R})$$

for the projective Veech group, and

$$\Gamma^\pm = \mathbb{P}D\,\mathrm{Aff}^\pm(X, \omega) \subset \mathrm{Isom}(\mathbb{H})$$

for its extended projective Veech group. Here projectivization removes the scalar ambiguity in passing from derivative groups to isometries of the hyperbolic plane.

Let e be an edge. The derivative of the edge reflection has two eigendirections: the $+1$-eigendirection θ_e, parallel to e, and the -1-eigendirection $\theta_e^{\perp}$, perpendicular to e. By Theorem **??**, θ_e is parabolic. By the preceding remark and the Veech dichotomy, $\theta_e^{\perp}$ is parabolic as well. Denote the corresponding two parabolic fixed points by

$$\xi_e^+, \xi_e^- \in \partial\mathbb{H}.$$

The edge reflection σ_e gives an orientation-reversing element

$$R_e = \mathbb{P}D\sigma_e \in \Gamma^{\pm}.$$

The fixed points of the induced projective action on $\partial\mathbb{H} \cong \mathbb{RP}^1$ are precisely the two eigendirections of $D\sigma_e$. Hence its hyperbolic mirror has ideal endpoints ξ_e^+ and ξ_e^-.

In particular, the hyperbolic mirror of an edge reflection is a cusp-to-cusp geodesic in the Teichmüller disk. Its two endpoints may represent the same cusp after passage to the quotient $\mathbb{H}/\Gamma$.

Proposition 5.1 (Extended stabilizer at a reflective endpoint)**.** *Let ξ be either endpoint ξ_e^+ or ξ_e^- of the hyperbolic mirror of R_e, and let*

$$\Gamma_\xi = \langle T \rangle$$

be its orientation-preserving stabilizer, with T a primitive parabolic generator. Then

$$R_e T R_e^{-1} = T^{-1},$$

and

$$(\Gamma^{\pm})_\xi = \langle T, R_e \rangle \cong D_\infty.$$

Proof. Conjugate in $\mathrm{PGL}_2(\mathbb{R})$ so that $\xi = \infty$ and the hyperbolic mirror of R_e is the imaginary axis. Then

$$T(z) = z + \lambda, \qquad R_e(z) = -z$$

for some $\lambda > 0$. Hence

$$R_e T R_e^{-1}(z) = z - \lambda = T^{-1}(z).$$

Thus $\langle T, R_e \rangle \cong \mathbb{Z} \rtimes_{-1} \mathbb{Z}/2 = D_\infty$.

If $S \in (\Gamma^{\pm})_\xi$ is orientation preserving, then $S \in \Gamma_\xi = \langle T \rangle$. If S is orientation reversing, then SR_e is orientation preserving and fixes ξ, so $SR_e = T^m$ for some $m \in \mathbb{Z}$. Thus $S = T^m R_e$, proving that the displayed subgroup is the full stabilizer. □

Corollary 5.2 (The real mirror joins cusps)**.** *The reflection R_e induces an antiholomorphic involution of the Teichmüller orbifold*

$$C_P = \mathbb{H}/\Gamma.$$

The image of its hyperbolic mirror lies in the fixed locus of this involution and joins the cusps represented by ξ_e^+ and ξ_e^-; the two ends are allowed to represent the same cusp of C_P.

Proof. The orientation-preserving subgroup is normal in the extended group, so $R_e \Gamma R_e^{-1} = \Gamma$. Therefore R_e descends to an antiholomorphic involution of $\mathbb{H}/\Gamma$. Its mirror is fixed pointwise, and its two ideal endpoints are the parabolic points ξ_e^+ and ξ_e^-. □

Thus the reflection that detects the edge direction on the flat surface remains visible as a real cusp-to-cusp geodesic on the Teichmüller orbifold.

6 Dihedral edge classes and examples

In this section an *edge cusp* means the cusp represented by the edge-parallel direction ξ_e^+; the perpendicular endpoint of the reflection mirror is not counted in this terminology.

We use the convention that D_N has order $2N$. All Veech groups in this section are projective; thus, for example, the regular-octagon Veech orbifold has signature $(4, \infty, \infty)$ in the projective convention [**?**, Theorem 5.4]. The corresponding nonprojective derivative-group convention retains the central $-I$ and is sometimes written with an elliptic order 8.

Proposition 6.1 (Edge-cusp classes)**.** *Suppose the directional reflection group of a lattice polygon satisfies*

$$G(P) \cong D_N,$$

and let

$$G^+(P) = G(P) \cap \mathrm{SO}(2) \cong C_N.$$

The edge directions form one $G^+(P)$-orbit if N is odd and exactly two $G^+(P)$-orbits if N is even. Consequently the edge cusps form

$$\begin{cases} 1, & \text{if } N \text{ is odd,} \\ 1 \text{ or } 2, & \text{if } N \text{ is even,} \end{cases} \qquad \textit{Veech-group orbits.}$$

Proof. Choose generators

$$D_N = \langle \rho, s : \rho^N = s^2 = 1,\ s\rho s = \rho^{-1} \rangle.$$

The reflections are $\rho^k s$, $k \in \mathbb{Z}/N$, and conjugation by the rotation subgroup satisfies

$$\rho^j (\rho^k s) \rho^{-j} = \rho^{k+2j} s.$$

If N is odd, multiplication by 2 is invertible modulo N, so all reflections form one C_N-conjugacy class.

If N is even, there are two conjugacy classes, according to the parity of k. Both classes occur among actual edge reflections. Indeed, if all edge reflections had the same parity, then the product of any two would lie in $\langle \rho^2 \rangle$, so the subgroup generated by the edge reflections would have rotational subgroup contained in the proper subgroup $\langle \rho^2 \rangle < \langle \rho \rangle$. This contradicts the definition of $G(P)$ as the group generated by the edge reflections.

Finally $G^+(P)$ acts on the reduced unfolding, hence

$$G^+(P) \subset \Gamma.$$

Directions in one $G^+(P)$-orbit therefore determine one Veech cusp orbit. When N is even, additional affine symmetries may merge the two finite dihedral classes, so one or two Veech cusp orbits can occur. □

The even case already exhibits both the finite-dihedral splitting and the possibility of further merger by the full affine group.

Example 6.2 (The regular octagon). Let O be a regular octagon regarded as a billiard table. Its edge-direction reflection group is

$$G(O) \cong D_4.$$

Since the centered regular octagon is preserved by every element of $G(O)$, its translation-redundancy subgroup is $H = G(O)$. Thus the reduced unfolding consists of a single regular octagon, with opposite sides identified.

The edge directions form two orbits under the rotation subgroup $G^+(O) \cong C_4$. On this reduced regular-octagon surface, however, rotation through $\pi/4$ is an affine automorphism [**?**, §5]. It exchanges the two $G^+(O)$-orbits. Consequently all actual edges of the regular octagon represent a single Veech cusp orbit.

The projective Veech group is the $(4, \infty, \infty)$ triangle group [**?**, Theorem 5.4], so the Teichmüller orbifold has two cusp orbits; see also [**?**, §6.2.2]. Since all actual octagon edges lie in the single orbit just described, the second cusp orbit is not represented by an actual edge direction of this billiard presentation.

Example 6.3 (The $\pi/8$ right triangle). Let T be the right triangle with angles

$$\left(\frac{\pi}{8}, \frac{3\pi}{8}, \frac{\pi}{2}\right).$$

The triangle is scalene, hence its translation-redundancy subgroup H is trivial. Thus its full and reduced unfoldings coincide. The classical unfolding glues sixteen copies of T into a regular octagon with opposite sides identified [**?**, §1.5]. Choosing the legs to be horizontal and vertical, the three edge directions modulo π are

$$0, \qquad \frac{\pi}{2}, \qquad \frac{7\pi}{8}.$$

The least common multiple of the angle denominators is 8, so

$$G(T) \cong D_8$$

in our convention [**?**, §1.5]. The first two edge reflections lie in one $G^+(T)$-class and the hypotenuse reflection lies in the other.

The projective Veech group of the regular-octagon surface is the $(4, \infty, \infty)$ triangle group [**?**, Theorem 5.4]. Its two cusp orbits may be represented by the boundary points

$$\infty \qquad \text{and} \qquad \pm \cot \frac{\pi}{8}$$

in the upper-half-plane model [**?**, §6.2.2]. Under the direction-to-boundary convention used for the Teichmüller disk, a Euclidean direction of slope m corresponds to the boundary point $-1/m$; see [**?**, §6.1.3]. Hence the horizontal direction represents the cusp at ∞, while

$$-\frac{1}{\tan(7\pi/8)} = \cot \frac{\pi}{8}$$

shows that the hypotenuse direction represents the other cusp. Therefore

$$[0]_\Gamma = [\pi/2]_\Gamma \neq [7\pi/8]_\Gamma = [3\pi/8]_\Gamma.$$

This agrees with the explicit cylinder decompositions of Gutkin–Hubert–Schmidt, who identify the coordinate directions as one parabolic pair and $3\pi/8, 7\pi/8$ as another [**?**, Example 4]. Massart's comparison of the two cylinder-modulus ratios gives an independent geometric way to distinguish the two cusps [**?**, §6.2.2].

Thus the two finite-dihedral edge classes remain two distinct Veech cusp classes for this billiard presentation.

Examples 6.2 and 6.3 have translation-isomorphic reduced unfoldings but different subsets of cusp classes represented by actual edges. In this sense, the edge-cusp data remembers the billiard presentation, not only the underlying Veech surface.

Remark 6.4 (Scope). The main argument concerns polygonal billiard tables equipped with the standard reflection unfolding and its translation-redundancy reduction. It also applies verbatim to any setting in which the same equivariant full-copy construction and quotient are available. No assertion is intended for an arbitrary polygonal quotient of a Veech surface lacking such a reflection model.

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Independent review and Lean verification notice

No independent human mathematical review has been performed.

A related strict Lean formalization of McMullen's Question 3.13 is under active development. This GitHub-public release does not claim that the full natural-language argument in this paper has already been completely formalized, nor does it assign an F1 or F2 verification label. The working local verification convention is to compile the selected development as `McMullen313.lean` using `lake env lean McMullen313.lean`. Any local build and kernel execution reported for this project is Chuyang Chen's run, not a joint or collaborative execution record.

Successful compilation of an intermediate Lean file, when recorded separately, establishes only the kernel acceptance of that file in the stated environment. It does not by itself establish complete formalization of every assertion in this paper or exact correspondence with McMullen's original Question 3.13.

Provenance and contacts

Project curator. Chuyang Chen.

AI-generation coordinator and contact. Chuyang Chen (22535018@zju.edu.cn).

Mathematical draft generation. Mathematical drafts were generated and revised with GPT-5.6 Sol under Chuyang Chen's supervision.

Lean code generation. The related Lean source has been generated and revised with GPT-5.6 Sol under Chuyang Chen's supervision. This is a code-generation provenance statement and is separate from mathematical review.

Lean build/kernel execution. Local build and kernel-check commands for this project are executed by Chuyang Chen. This is not a collaborative run record.

Human mathematical verification. NONE.

Statement-correspondence audit. NONE. No independent human audit has certified that the natural-language statement and any current formal Lean statement are exactly equivalent.

Comments, corrections, historical references, and independent verification are very welcome. For long-term correspondence, readers are encouraged to use the repository issue tracker as well as the contact above.

Lean formal verification record

This record separates the natural-language statement, the developing formal Lean statement, kernel checking of individual development versions, and the separate audit of the correspondence between the first two. No completed formal-verification label is asserted here.

Formalization status. CONDITIONAL FORMALIZATION – the algebraic and combinatorial core is verified in Lean, while several geometric ingredients are represented as explicit external inputs.

Working source convention. The main source file is `McMullen313.lean`; the repository contains the corresponding auxiliary files and verification records.

Build command. `lake env lean McMullen313.lean`.

Kernel verification scope. The formalized development contains no hidden proof placeholders such as `sorry`; the remaining assumptions appear as explicit hypotheses of the conditional theorems.

External inputs. The current formalization isolates the leafwise Veech-type input, the finite-obstruction periodicity input, and the horizontal parabolic-cylinder input as separate mathematical statements to be supplied independently.

Scope note. The verified Lean theorem establishes the implication from these explicit geometric hypotheses to the stated periodicity conclusions. It does not claim that the external geometric inputs have already been formalized inside Lean.

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